Unit of Observation

Each row in this dataset represents a single trading day for Amazon (AMZN) stock, containing information about its price movements and trading activity on that specific date.

Dataset Summary

- Total observations: ~7,000+ daily trading records (1997–2026)
- Data type: Time series financial data
- Source: MacroTrends Amazon dataset
- Frequency: Daily observations
- Variables include:
    - Date
    - Open, High, Low, Close
    - Volume
    - Log-transformed variables (used for modeling)

Variable Descriptions

1. Date

- Type: Datetime
- Description: The trading date for the recorded stock data
- Notes:
    - Serves as the time index for the dataset
    - Used to order observations chronologically for time series modeling
    - No missing values expected

2. Open

- Type: Float
- Description: The opening price of Amazon stock on a given trading day (USD)
- Notes:
    - Represents the first traded price of the day
    - Typically close to the previous day's closing price unless there is a market shock

3. High

- Type: Float
- Description: The highest price reached during the trading day (USD)
- Notes:
    - Captures intraday peak price

    - Useful for measuring volatility

4. Low

- Type: Float
- Description: The lowest price reached during the trading day (USD)
- Notes:
    - Captures intraday minimum price
    - Used alongside “High” to understand price range

5. Close

- Type: Float
- Description: The closing price of Amazon stock at the end of the trading day (USD)
- Notes:
    - Primary variable used for forecasting in this project
    - Reflects final market valuation for the day
    - Subject to structural breaks (e.g., 2022 stock split)

6. Adj Close (if included in your dataset)

- Type: Float
- Description: Adjusted closing price accounting for stock splits and dividends
- Notes:
    - More consistent across time compared to raw Close
    - Considered during preprocessing due to 2022 stock split

7. Volume

- Type: Integer
- Description: The number of shares traded during the day
- Notes:
    - Indicates market activity and liquidity
    - Often highly skewed and may require transformation

8. Log-Transformed Variables

log_close

- Type: Float

- Description: Log-transformed closing price
- Purpose:
    - Reduces skewness
    - Stabilizes variance for modeling

log10_vol

- Type: Float
- Description: Log-transformed trading volume
- Purpose:
    - Normalizes highly skewed volume data

Exploratory Plots and Tables

Figure 1: Time series line plot of Amazon's daily closing price (1997–2026), showing the full historical trend with key events annotated (dot-com crash, 2008 crisis, COVID-19 dip, stock split).

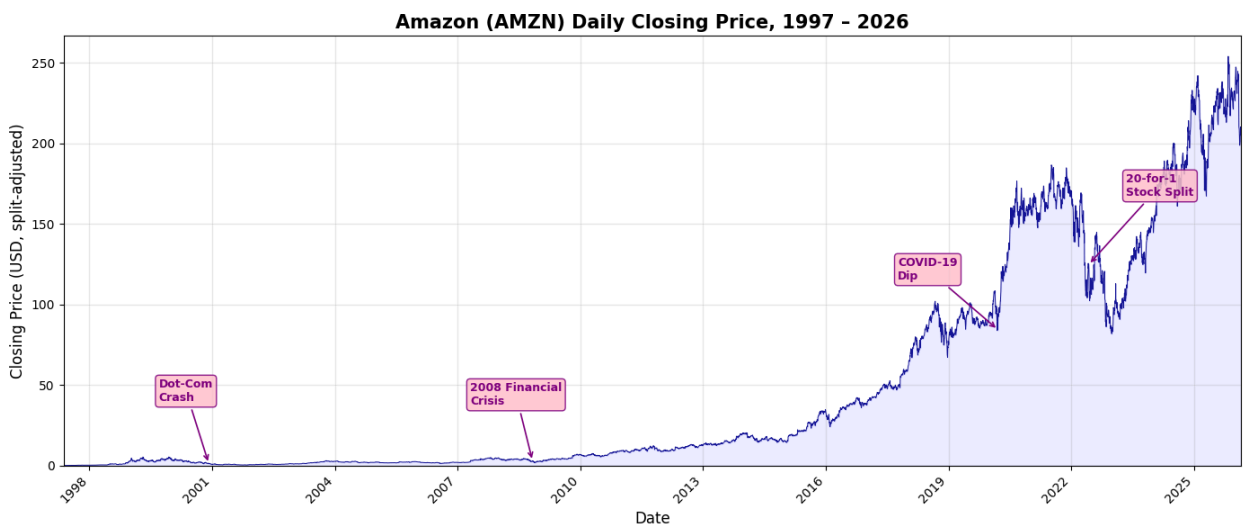


Figure 2: Histogram and/or density plot of the closing price distribution, demonstrating the right-skewed nature of the data. A second panel shows the log-transformed distribution for comparison.

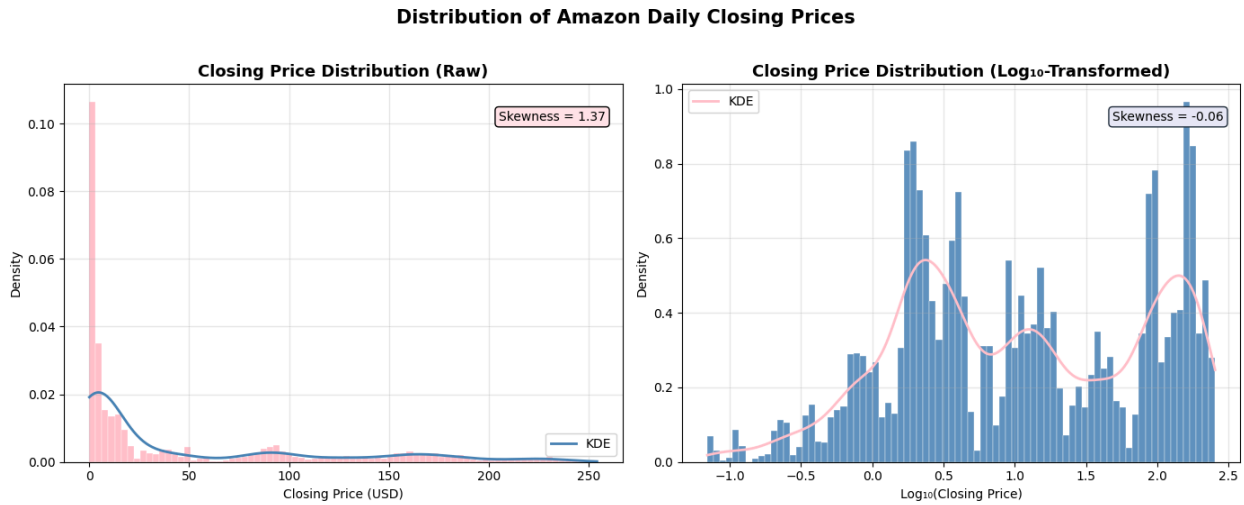
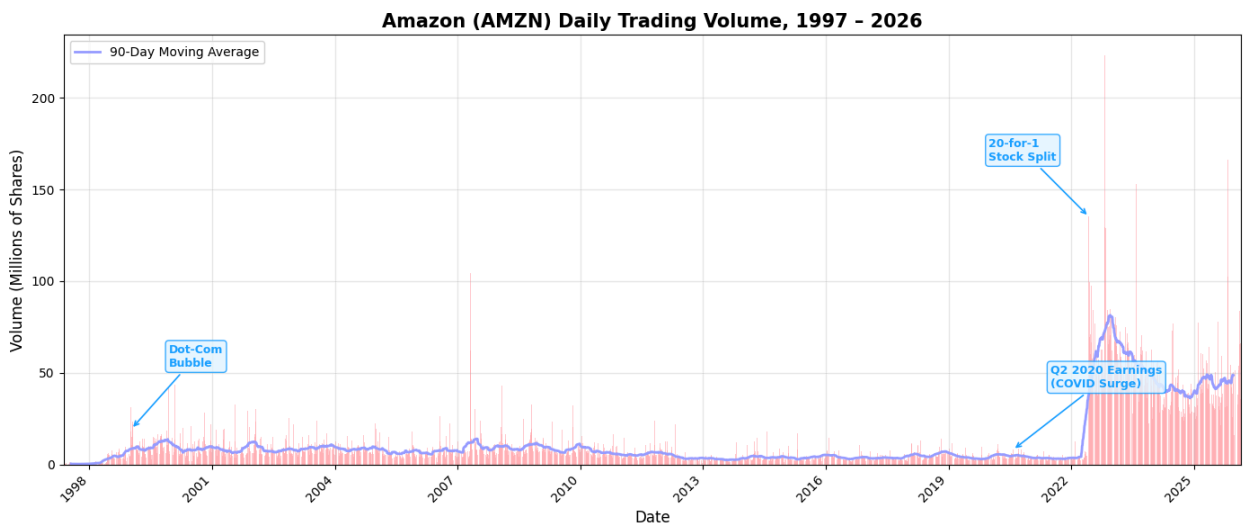


Figure 3: Volume over time plot, showing trading volume trends (90-day rolling average) and notable spikes corresponding to major market events.



Model Output & Evaluation

Figure 1: This figure displays the ACF and PACF plots of the differenced closing price, showing that most autocorrelation values fall within the confidence bounds, which suggests the series is approximately stationary with minimal remaining autocorrelation.

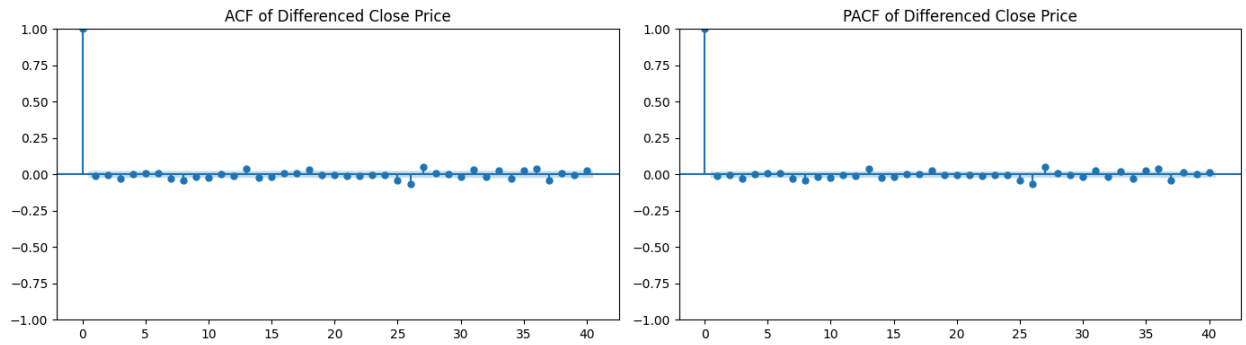


Figure 2: This shows the time series of Amazon's closing price split into training, validation, and test sets, illustrating how the data is partitioned chronologically for model development and evaluation.

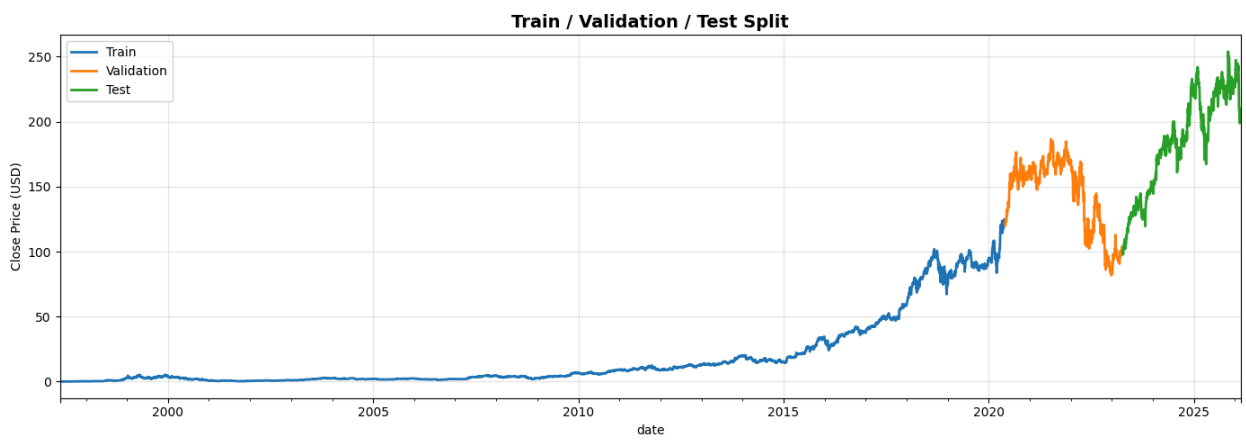
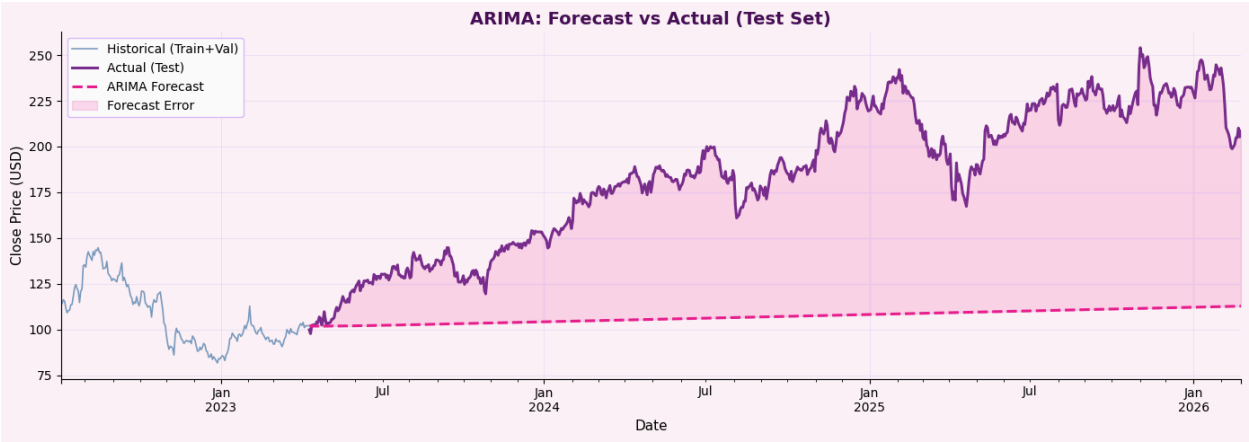


Figure 3: This figure shows the ARIMA model forecast compared to the actual closing prices on the test set, highlighting the difference between predicted and observed values and the resulting forecast error over time.


ARIMA: Forecast vs Actual (Test Set)
Historical (Train+Val)
Actual (Test)
ARIMA Forecast
Forecast Error
Close Price (USD)
250
225
200
175
150
125
100
75
Jan 2023
Jul
Jan 2024
Jul
Jan 2025
Jul
Jan 2026
Date